



Lecture 14

Chance

Announcements

- HW 5 due Wed, 2/21
 - Project 1 due next Friday at 5pm
 - Checkpoint due today at 5pm
 - Midterm on Friday, March 8, 7-9pm PT
 - Scope: everything up to and including lecture that day
 - Fill out [Accommodations Form](#) by **Feb 24** at 11:59pm!
 - No OH or lecture on Monday!
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Random Selection

Random Selection

`np.random.choice`

- Selects uniformly at random
- with replacement
- from an array,
- a specified number of times

```
np.random.choice(some_array, sample_size)
```

Appending Arrays

A Longer Array

- **`np.append(array_1, value)`**
 - new array with `value` appended to `array_1`
 - `value` has to be of the same type as elements of `array_1`
 - **`np.append(array_1, array_2)`**
 - new array with `array_2` appended to `array_1`
 - `array_2` elements must have the same type as `array_1` elements
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Iteration

for Statements

- **for** is a keyword that begins a multiline **for** statement.
 - Executing a **for** statement performs a computation for every element in a list or array.
 - A common special case is to perform a computation a fixed number of times.
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Anatomy of a for loop

Example:

```
variable name      array of values
for item in some_array:
indent     print(item)
code to evaluate in each iteration of for loop
```

Simulation

(Demo)

Chance

Basics

- **Lowest value:** 0
 - Chance of event that is impossible
 - **Highest value:** 1 (or 100%)
 - Chance of event that is certain
 - **Complement:** If an event has chance 70%, then the chance that it doesn't happen is
 - $100\% - 70\% = 30\%$
 - $1 - 0.7 = 0.3$
-

Equally Likely Outcomes

Assuming all outcomes are equally likely, the chance of an event A is:

$$P(A) = \frac{\text{number of outcomes that make A happen}}{\text{total number of outcomes}}$$

A Question

- I have three cards: **ace of hearts**, **king of diamonds**, and **queen of spades**.
- I shuffle them and draw two cards *at random without replacement*.
- What is the chance that I get the Queen followed by the King?

(Demo)

Multiplication Rule

Chance that two events A and B both happen

= $P(A \text{ happens}) \times P(B \text{ happens given that } A \text{ has happened})$

- The answer is *less than or equal to* each of the two chances being multiplied
 - The more conditions you have to satisfy, the less likely you are to satisfy them all
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Another Question

- I have three cards: **ace of hearts**, **king of diamonds**, and **queen of spades**.
- I shuffle them and draw two cards *at random without replacement*.
- What is the chance that one of the cards I draw is a King and the other is Queen?

(Demo)

Addition Rule

If event A can happen in *exactly one* of two ways, then

$$P(A) = P(\text{first way}) + P(\text{second way})$$

- The answer is *greater than or equal to* the chance of each individual way
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Complement: At Least One Head

- In 3 tosses:
 - Any outcome *except* TTT
 - $P(\text{TTT}) = (1/2) \times (1/2) \times (1/2) = 1/8$
 - $P(\text{at least one head}) = 1 - P(\text{TTT}) = 1 - (1/8) = 87.5\%$
- In 10 tosses:
 - $1 - (1/2)^{10} \approx 99.9\%$

(Demo)

Problem-Solving Method

Here's a method that works widely:

Ask yourself what event must happen on the first trial.

- If there's a clear answer (e.g. “not a six”) whose probability you know, you can most likely use the **multiplication rule**.
 - If there's no clear answer (e.g. “could be K or Q, but then the next one would have to be Q or K ...”), list all the **distinct ways** your event could occur and **add up their chances**.
 - If the list above is long and complicated, look at the **complement**. If the complement is simpler (e.g. the complement of “at least one” is “none”), you can find its chance and subtract that from 1.
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Discussion Question

A population has 100 people, including Kendall and Roman.
We sample two people at random without replacement.

(a) $P(\text{both Kendall and Roman are in the sample})$

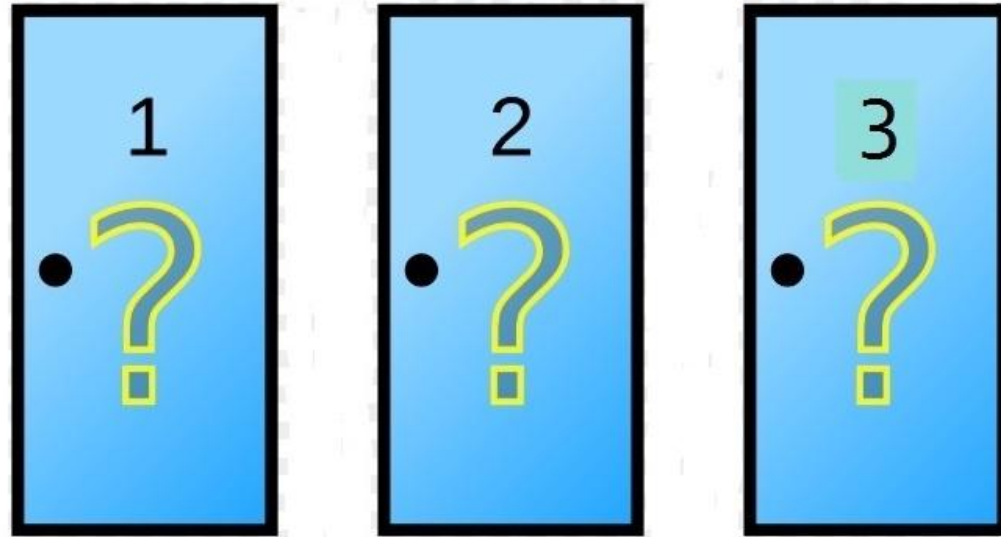
$$\begin{aligned} &= P(\text{first Kendall, then Roman}) + P(\text{first Roman, then Kendall}) \\ &= (1/100) * (1/99) + (1/100) * (1/99) = 0.0002 \end{aligned}$$

(b) $P(\text{neither Kendall nor Roman is in the sample})$

$$= (98/100) * (97/99) = 0.9602$$

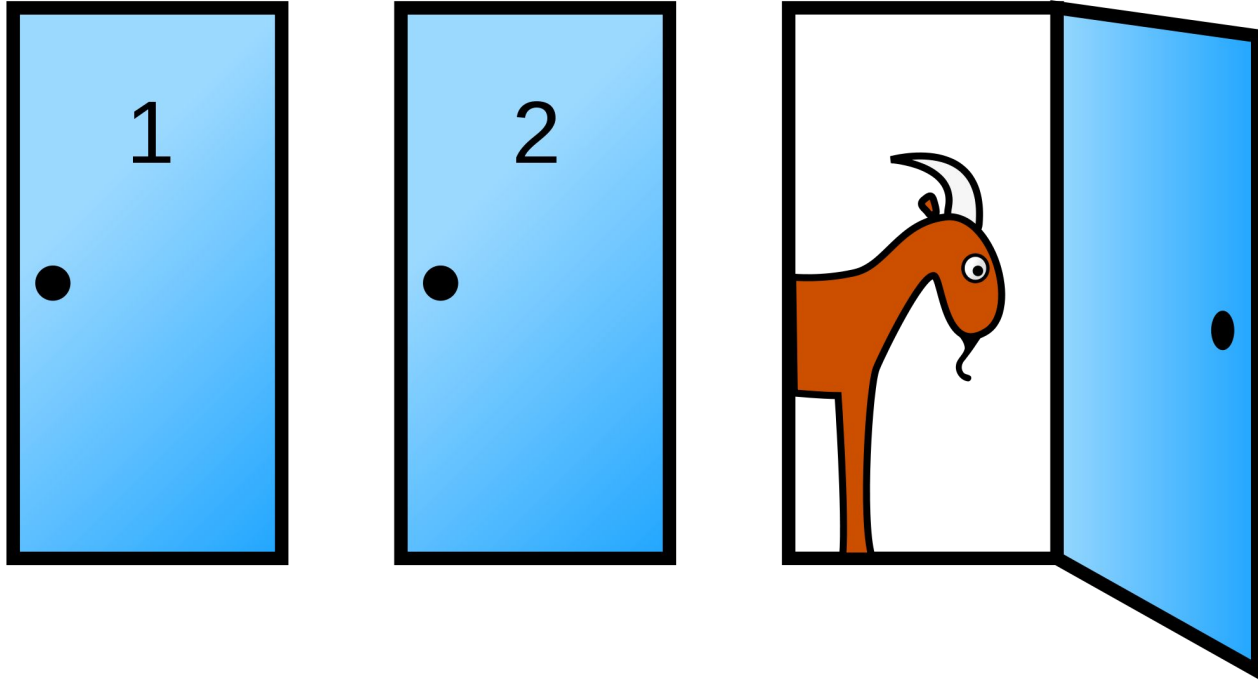
The Monty Hall Problem

Monty Hall Problem



<https://probabilityandstats.files.wordpress.com/2017/05/monty-hall-pic-1.jpg>

Stay or Switch?



https://en.wikipedia.org/wiki/Monty_Hall_problem

Equally Likely Outcomes

My Choice

 Adhikari

 DeNero



Monty's Reveal

 DeNero

 Adhikari

 Adhikari or  DeNero

Other Choice



 DeNero or  Adhikari